

Supplementary Material

Table S1: Multivariate regression analysis of factors associated with research visibility.

Ordinary least-squares regression was performed using $\log_{10}(\text{PubMed publication count} + 1)$ as the response variable. Predictors were standardized prior to analysis to facilitate comparison of effect sizes. Analyses were restricted to genes with complete phylostratum and expression information and detectable expression in at least one tissue ($n = 17,353$).

Predictor	Coefficient (β)	Std. error	t value	p value
Intercept	1.7054	0.0032	525.66	$< 2 \times 10^{-16}$
Phylostratum	-0.0822	0.0035	-23.60	$< 2 \times 10^{-16}$
Mean expression ($\log_{10}$)	0.2089	0.0035	59.56	$< 2 \times 10^{-16}$
Tissue specificity (τ)	-0.0148	0.0036	-4.16	3.2×10^{-5}

Model statistics: $R^2 = 0.256$, adjusted $R^2 = 0.256$, F-statistic = 1987, $p < 2 \times 10^{-16}$. Higher phylostratum values correspond to younger inferred gene ages.

Table S2: Conceptual contrast between highly studied genes and low-visibility genes.

This table summarizes the principal qualitative differences between the two most contrasting research-attention strata (top500 and randomBelow10k) across evolutionary, molecular, and functional dimensions, integrating observations from Figs. 2-5. Intermediate groups are not shown for clarity.

Feature	Top500	RandomBelow10k
Research attention	Highly studied; dominant in literature	Low visibility; below rank 10,000
Evolutionary constraint	Stronger long-term constraint	Moderately relaxed constraint; heavier upper tail
Gene age	Intermediate to older profile	Enriched for younger, lineage-restricted genes
Expression profile	Higher expression; broader across tissues	Lower expression; increased tissue specificity
Sequence features	Narrow GC-content regime; not depleted of long CDSs	Broader GC distribution; increased density of shorter CDSs
Experimental tractability	Strong cross-species homology; amenable to model systems	Reduced homology; context-dependent expression
Methodological implication	Well captured by standard comparative and network-based approaches	Likely underdetected by standard pipelines; requires context-specific and human-relevant systems
High-level interpretation	Conserved, constrained genes with extensive functional characterization	Distinct class of younger, less constrained genes that remain systematically underexplored

Table S3: Loss of rank resolution across lower-visibility gene regions. Fixed 500-gene windows were evaluated across different regions of the PubMed publication-count ranking. Lower-ranked windows show increasingly compressed publication-count distributions, with fewer distinct count values, larger tie blocks, and more genes outside the focal window sharing identical publication counts. “Equivalent genes outside window” denotes genes excluded from the focal 500-gene window despite having publication counts identical to genes within that window.

Window	Rank start	Rank end	Min count	Max count	Range	Unique counts	Max tie size	Equivalent genes outside window
top500	1	500	518	11,608	11,090	383	4	0
ranks_4501_5000	4,501	5,000	97	107	10	11	60	50
ranks_9501_10000	9,501	10,000	43	47	4	5	148	225
ranks_14501_15000	14,501	15,000	20	22	2	3	281	328
ranks_19501_20000	19,501	20,000	7	8	1	2	376	525

Table S4: Public data sources used in this study. For each resource, the table lists the dataset or file used, its role in the analysis, the source repository/database, and a direct web link.

Analysis component	Dataset / file	Repository / database	Identifier / version	Direct link
Gene model definition / representative transcript selection	MANE Select transcript set	Matched Annotation from NCBI and EMBL-EBI (MANE)	MANE GRCh38 v1.5 (MANE.GRCh38.v1.5.summary.txt)	https://www.ncbi.nlm.nih.gov/refseq/MANE/
Transcript and protein sequence retrieval	RefSeq mRNA and protein sequences corresponding to MANE Select transcripts	NCBI Reference Sequence Database (RefSeq)	See processed table	https://www.ncbi.nlm.nih.gov/refseq/
Gene-level publication ranking	Gene-PubMed publication count mapping (gene2pubmed.gz)	NCBI Gene / PubMed	Downloaded 2025; includes all Gene IDs linked to PubMed (no symbol-based aggregation)	https://ftp.ncbi.nlm.nih.gov/gene/DATA/
Expression analysis	rna_tissue_consensus.tsv	The Human Protein Atlas	HPA v24 (downloaded 18 Feb 2026)	https://www.proteinatlas.org/about/download
Disease-gene association analysis	human_disease_knowledge_filtered.tsv; human_disease_knowledge_full.tsv	DISEASES database (Jensen-Lab)	Downloaded 2025 (files: filtered and full knowledge datasets)	https://diseases.jensenlab.org/Downloads
Disease resource landing page	Disease-gene association database	DISEASES database (Jensen-Lab)	Not applicable	https://diseases.jensenlab.org/
Rare disease / orphan classification	Orphanet classification files	Orphadata / Orphanet	Dec 2025 release (downloaded 10 Dec 2025)	https://www.orphadata.com/_classifications/
Rare disease classification landing page	Rare disease classification browser	Orphanet	Not applicable	https://www.orpha.net/en/disease/classification

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Analysis component	Dataset / file	Repository / database	Identifier / version	Direct link
Evolutionary rate analysis	Human–rhesus macaque alignments (Kosiol et al.)	Mammalian positive selection dataset (Kosiol et al.)	As reported in [13] (accessed 2025)	http://compgen.bscb.cornell.edu/projects/mammal-psg/
Gene age analysis	Phylostratum assignments	Published dataset (Litman and Stein)	As reported in [14] (Supplementary dataset)	See supplementary materials of cited publication
Code, processed data, and derived tables	Analysis scripts and processed datasets	Private repository (review access)	Review-access repository	https://tu-dortmund.sciebo.de/s/GroQkLZNFWZTG8H